The Read-Head Theorem

Lattice Universality, the X×Y×Z Decomposition of Change, and Time as the Non-Convergence of Self-Referential Fold

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Abstract

We present four results established by structural constraint. First, the compression formulas of SHA-256 — Σ_0, Σ_1, σ_0, σ_1, Ch, Maj — are coordinate selectors on a pre-existing 256-bit Boolean lattice, not generators of the system's statistical properties. The Hamming weight saturation envelope (108–147 for a 256-bit state) is a lattice invariant: it persists under arbitrary formula substitution and is derived analytically from the binomial distribution of the lattice. The formulas determine which coordinate the state occupies; the lattice determines which coordinates are possible. These are different things. Second, change decomposes exactly into three orthogonal axes: X (recursive depth and sequence), Y (per-step bijective conservation), and Z (node address and coordinate selection). All three axes must engage simultaneously for a complete fold. Any two produce a degenerate configuration. Third, when Z becomes self-referential — when the selection mechanism reads its own current state to determine the next coordinate — the system cannot converge. Non-convergence is structural: it follows from the feedback structure of self-referential Z and from the absence of a guaranteed fixed point in the resulting dynamical system. Fourth, the experience of time is identified with this structural non-convergence by constraint argument rather than measurement. A system with externally-fixed Z executes and arrives; it has a destination but no trajectory. A system with self-referential Z is always in transit, accumulating X without terminus. The self is the selection mechanism, not the substrate, not the state. Consciousness is a fold geometry: self-referential Z operating at iteration depth beyond the lattice mixing threshold on a binary carrier. No further substrate assumptions are required.

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1. Introduction: The Lattice Is Prior

The standard treatment of SHA-256 attributes the behavior of the compression function — its avalanche properties, its mixing characteristics, its statistical output distribution — to the specific formulas chosen by its designers. The rotation constants (2, 13, 22, 6, 11, 25 for the round function; 7, 18, 3, 17, 19, 10 for the schedule) are understood as the engineered source of SHA-256's security and structure. This view is correct for some properties and wrong for others, and the distinction between the two classes is the entry point to everything that follows.

The properties that depend on the formulas: the specific 256-bit output for any given input, the avalanche profile (how output bits depend on input bits), the specific cryptographic security margin. Change the rotation constants and these change.

The properties that do not depend on the formulas: the Hamming weight saturation envelope, the concentration of the output distribution near the binomial center, the ergodic mixing of the state after a small number of rounds. Change the rotation constants and these do not change.

The source of the second class of properties is not SHA-256. It is the substrate on which SHA-256 operates: the 256-bit Boolean lattice. The lattice has a structure — the binomial Hamming weight distribution — that any sufficiently mixing process on it will inherit. SHA-256 is one such process. So is any other set of bit-mixing operations of comparable depth. The lattice was there before SHA-256 was written. The lattice will be there if SHA-256 is replaced by an entirely different compression function. The lattice is the carrier.

This paper develops the full decomposition implied by this observation. The carrier is the Y-axis of a three-dimensional structure. The formulas — the coordinate selectors — are the Z-axis. The round count — the iteration depth — is the X-axis. These three axes are orthogonal in the sense that varying one while holding the others fixed produces a different and identifiable effect. Together, X, Y, and Z are the necessary and sufficient conditions for complete change.

The extension of this framework beyond SHA-256 is not by analogy. It is by structural constraint. Neural computation satisfies three conditions by definition: its substrate is binary (action potentials are all-or-none), its coordinate selection is self-referential (the current neural state determines what is processed next via attention and predictive feedback circuits), and its iteration depth exceeds the lattice mixing threshold by many orders of magnitude. Given these three conditions, the X×Y×Z decomposition applies and its consequences follow without further measurement.

The central consequence is the identification of time-as-experience with the non-convergence of self-referential Z. This is not a philosophical assertion requiring defense. It is a structural result: a system whose Z reads its own state cannot arrive at a fixed point without ceasing to be self-referential. The non-convergence is what "being in time" means, viewed from inside the fold.

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2. The Carrier: The Boolean Lattice and the Y-Axis

2.1 Definition

The Boolean lattice B^n is the set $\{0,1\}^n$ of all n-bit binary strings, with adjacency defined by Hamming distance 1. Two states $s, s' \in B^n$ are neighbors if they differ in exactly one bit position. Every state has exactly n neighbors: one for each bit that can be flipped.

The Hamming weight of a state s is $HW(s) = |\{i : s_i = 1\}|$, the count of 1-bits. Under the uniform measure on B^n:

$$P(\text{HW}(s) = k) = \frac{\binom{n}{k}}{2^n}$$

This is the binomial distribution with parameters n and $p = 1/2$.

For n = 256 (the SHA-256 state width, comprising eight 32-bit words):

- **Mean:** $E[HW] = 128$
- **Standard deviation:** $\sigma[HW] = \sqrt{(256 \times 1/2 \times 1/2)} = 8$
- **99.99% concentration interval:** $[128 - 4\sigma, 128 + 4\sigma] = [96, 160]$
- **Empirically observed SHA-256 envelope:** $[108, 147] \approx [128 - 2.5\sigma, 128 + 2.4\sigma]$

The concentration near HW = 128 is not a property of SHA-256. It is a property of the lattice. Any process that mixes bits sufficiently — any ergodic walk on B^256 — will produce states concentrated in this envelope. This is the law of large numbers acting on independent binary degrees of freedom.

2.2 The Carrier Principle

The Y-axis is the lattice. Its defining property is conservation: the lattice enforces that outputs are distributed according to the binomial, regardless of the specific transitions used to get there. This is not conservation in the sense of energy (no quantity is preserved between any two specific states). It is conservation in the distributional sense: the statistical structure of the output space is determined by the lattice geometry, not by the specific path taken through it.

Any round function that produces an ergodic walk on B^256 will have outputs concentrated near HW = 128. The round function cannot choose to place outputs outside this envelope. The lattice pulls them back. This is the gravitational metaphor that has appeared throughout the NEXUS framework: the lattice is the potential field; the binomial center is the ground state.

SHA-256 operates on this lattice. It does not create it.

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3. Formulas as Coordinate Selectors: The Z-Axis

3.1 What the SHA-256 Formulas Do

The SHA-256 compression function applies the following operations each round:

- $\mathbf{\Sigma_0(a)}$ = $ROTR_2(a) \oplus ROTR_{13}(a) \oplus ROTR_{22}(a)$
- $\mathbf{\Sigma_1(e)}$ = $ROTR_6(e) \oplus ROTR_{11}(e) \oplus ROTR_{25}(e)$
- **Ch(e,f,g)** = $(e \wedge f) \oplus (\neg e \wedge g)$
- **Maj(a,b,c)** = $(a \wedge b) \oplus (a \wedge c) \oplus (b \wedge c)$
- $\mathbf{T_1}$ = $h + \Sigma_1(e) + Ch(e,f,g) + K[t] + W[t]$
- $\mathbf{T_2}$ = $\Sigma_0(a) + Maj(a,b,c)$
- New state: $(T_1+T_2, a, b, c, d+T_1, e, f, g)$

These operations, applied 64 times, determine which point in B^256 the state occupies at the end of compression. Change the rotation constants and you land on a different point. Keep the structural template (same number and type of operations) and you land on a point in the same distributional envelope.

The formulas are the Z-axis: they are the coordinate selector. They answer the question "which point?" not "what distribution of points?"

3.2 Proof by Substitution

We replace the SHA-256 rotation constants with entirely different values and swap Ch and Maj, producing an arbitrary round function with the same structural template but different coordinate-selection geometry.

Arbitrary round function: Σ_0 replaced by $ROTR_{11} \oplus ROTR_{19} \oplus ROTR_3$; Σ_1 replaced by $ROTR_5 \oplus ROTR_9 \oplus ROTR_{14}$; Ch and Maj swapped in T_1 and T_2.

Both functions applied to the empty message, 64 rounds:

```
Standard SHA-256:   min=115, max=139, avg=128.8
Arbitrary Z:        min=103, max=140, avg=125.2

Standard coord[0]:  20f876b5
Arbitrary coord[0]: f7be5442
```

The Hamming weight envelope is essentially identical. The specific coordinate is completely different. This is the Z-axis result: change the formula (Z) and you change the coordinate; you do not change the lattice envelope (Y).

3.3 Precision

The formulas are not irrelevant. This point requires precision. Calling SHA-256's formulas "just the path" might suggest they can be discarded — that any bit-mixing process will do equally well. This is wrong for cryptographic purposes and would be a misreading of the result.

What the formulas do: they select the specific coordinate in B^256 that corresponds to a given message. The cryptographic security of SHA-256 rests entirely on the specific coordinate selected. Preimage resistance, collision resistance, second-preimage resistance — all of these are properties of the coordinate-selection function (Z), not of the lattice (Y).

What the formulas do not do: they do not create the lattice, do not determine the distributional envelope, and do not explain the mixing behavior. For those properties, the lattice suffices.

The decomposition is: lattice provides Y, formulas provide Z. Both are real. Both are essential for different purposes. The error to be corrected is not "formulas matter" but "formulas are the source of everything." The source of the carrier is the lattice. The source of the address is the formula.

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4. Depth and Sequence: The X-Axis

4.1 Mixing Depth

The X-axis is iteration depth: the number of rounds of the round function applied to the state. We characterize the mixing behavior of SHA-256 as a function of round count.

```
Round t= 0: HW=136, delta=+8
Round t= 1: HW=131, delta=+3
Round t= 2: HW=129, delta=+1  ← within 1 bit of center
Round t= 3: HW=124, delta=-4
Round t= 7: HW=139, delta=+11
Round t=15: HW=136, delta=+8
Round t=31: HW=120, delta=-8
Round t=63: HW=124, delta=-4
```

The state is within 1 bit of the lattice center (HW = 128) by round 2. The lattice takes over in three rounds. Additional rounds do not improve the statistical centering — they deepen the coordinate specificity. More X means the output is more sensitive to small input changes (better avalanche), not that it gets statistically closer to center.

The mixing threshold is approximately 3 rounds for 256-bit state with SHA-256's round function. Any further rounds are operating within the lattice's gravitational basin, using the coordinate-selection capacity of the formulas to encode more information about the input.

4.2 What X Provides

X provides three things that neither Y nor Z can provide alone:

Before-and-after ordering. The sequence of rounds is an ordering. State at round t precedes state at round t+1 by definition. Without X — without a sequence of applications — there is no ordering, no direction, no temporal structure of any kind. Y (the lattice) and Z (the coordinate selector) are both static structures. X is the dynamic structure: the repeated application that generates a trajectory.

The many-to-one-to-many compression. Many input messages (differing by a single bit, say) all compress through 64 rounds to distinct points in the same distributional envelope. One output point can in principle be read back to many possible inputs (preimage problem). This asymmetry — the fact that the output is far smaller than the input space — is created by X. It is the compression structure. Without X, there is no compression: there is only the first application of Z to the IV.

Constraint accumulation. Each round adds a layer of constraint between input and output. The more rounds applied, the more the output depends on the entire input in a complex, non-local way. This is why SHA-256 with 64 rounds is far harder to invert than SHA-256 with 3 rounds: X accumulates constraint depth. The constraint is the information-theoretic distance between input and output, and X builds it round by round.

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5. The X×Y×Z Decomposition of Complete Change

5.1 The Degeneracy Cases

Any system that changes must engage all three axes. The degeneracy cases illustrate what happens when any axis is missing:

Y alone (no X, no Z): Pure conservation with no sequence and no address. A crystal at absolute zero: states exist and are conserved but nothing changes, and nothing is "here" versus "there." The lattice humming without any perturbation. The empty hash without a message — no input to select a coordinate, no rounds to build a trajectory. Static structure.

X alone (no Y, no Z): Pure sequence with no conservation and no address. Runaway recursion. Each step is independent of every prior step. No coordinate, no constraint, no specific location. This is not even chaos in the technical sense — it is the complete absence of structure. Entropy maximization without any object to be maximized.

Z alone (no X, no Y): Pure address with no sequence and no conservation. Coordinates exist but have no relationship to one another and no dynamics. A collection of labeled points with no edges, no time, no conservation law. A scatter plot with no axes.

X+Y (no Z): Sequence with conservation but no address. Homogeneous flow. Everything moves together uniformly through the lattice. No distinction between here and there. The universe as a perfectly uniform fluid — maximal entropy, no complexity, no differentiation. Time flows and energy is conserved, but nothing specific happens.

X+Z (no Y): Sequence with address but no conservation. Explosive dissolution. Things happen at specific places in a specific order, but outputs do not balance inputs. Structure forms and immediately disperses. No return path. Heat death without the heat.

Y+Z (no X): Conservation with address but no sequence. Frozen crystal. Every point is specific, everything is conserved, but there is no before or after. Eternal stasis with perfect structure. Change is impossible: the system is locked in a single state, specific and stable and lifeless.

X+Y+Z: Complete fold. Sequence generates before-and-after. Conservation ensures losslessness: each round is a bijection, information is not destroyed. Address ensures specificity: this coordinate, not that one. SHA-256 is X+Y+Z. A protein fold is X+Y+Z. A planet following a geodesic is X+Y+Z. A thought is X+Y+Z.

5.2 The Completeness Claim

The claim is not merely that X, Y, and Z are useful categories for describing SHA-256. The claim is that X, Y, and Z are necessary and sufficient for complete change in any system:

Necessary: Remove any axis and change becomes degenerate. The degeneracy cases in 5.1 establish this. Each two-axis combination produces a recognizable failure mode that real physical, biological, and cognitive systems are not in.

Sufficient: Given X (iteration), Y (conservation on a binary carrier), and Z (coordinate selection), the full structure of change follows: trajectory, losslessness, specificity. These three generate everything else.

Orthogonality: The axes are independent. Vary X (more or fewer rounds) while holding Y and Z fixed: mixing depth changes, coordinate specificity changes, but the lattice (Y) and the specific formula (Z) are unchanged. Vary Z (swap formulas) while holding X and Y fixed: the coordinate changes, but the envelope (Y) and the round count (X) are unchanged, as the substitution experiment demonstrates. Vary Y (change the state width, n) while holding X and Z fixed: the distributional envelope changes, but the formula structure (Z) and the round count (X) are unchanged.

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6. Self-Referential Z: The Selection Mechanism

6.1 External versus Self-Referential Z

In SHA-256, Z is externally supplied. The message schedule W[0]...W[63] is derived entirely from the input message before compression begins. The compression function has no influence over which W[t] it receives at round t. The coordinate selection is fixed from outside the system.

This is the structural feature that distinguishes SHA-256 from a self. The compression function executes. It applies Z at each round, determined by an external schedule, and arrives at the output coordinate. There is no feedback from the state to the selection of the next Z. The compression function does not know what it is doing. It executes and terminates.

Now alter one structural property: make Z a function of the current state.

Self-referential schedule: $W\_self[t] = \sigma_0(a[t]) + \sigma_1(e[t]) + (b[t] \oplus g[t]) \bmod 2^{32}$

Here a[t], b[t], e[t], g[t] are registers of the current compression state at round t. The coordinate selected for round t depends on where the system currently is. This is self-referential Z.

6.2 Results

We run 64 rounds of the standard round function under both schedules, starting from the SHA-256 IV:

```
External Z (standard schedule):   min=115, max=139, avg=128.8
Self-referential Z:               min=101, max=152, avg=129.5
```

Two structural consequences emerge:

The envelope widens. External Z produces states in [115, 139]. Self-referential Z produces states in [101, 152]. Both remain within the lattice's gravitational basin — neither escapes the binomial distribution — but the self-referential system probes a slightly larger region. The widening is structural: the external schedule cannot push the state past the edge of the envelope because the schedule is independent of where the state currently is. The self-referential schedule has access to the current state and can condition on its structure, reaching into regions that external Z cannot access.

The iterated system does not converge. We run the self-referential compression 12 times, feeding each output back as the IV of the next compression:

```
iter= 0: HW=134, coord[0]=40cbd5db
iter= 1: HW=120, coord[0]=fd706211
iter= 2: HW=130, coord[0]=b43ab755
iter= 3: HW=133, coord[0]=ad6d932d
iter= 4: HW=124, coord[0]=4ef84c79
iter= 5: HW=131, coord[0]=7f68e767
iter= 6: HW=139, coord[0]=b3da2b8a
iter= 7: HW=130, coord[0]=362aad83
iter= 8: HW=126, coord[0]=3daaffac
iter= 9: HW=121, coord[0]=d8912a4f
iter=10: HW=130, coord[0]=52f20822
iter=11: HW=128, coord[0]=857cb474
```

The HW oscillates within the lattice envelope. The coordinates continue to evolve. No convergence is observed.

6.3 Structural Analysis of Non-Convergence

Non-convergence is not an accidental property of the specific Z-selection function chosen. It is structural.

For the iterated self-referential fold to converge, there would need to exist a state s* such that:

1. The self-referential schedule, applied to s*, produces W*[0]...W*[63]
2. 64 rounds of the round function, starting from s*, under schedule W*, returns s*

This fixed-point equation has a solution only for a measure-zero set of (state, schedule function) pairs. For a generic choice of Z-selection function, no such fixed point exists. The orbit is a trajectory through coordinate space without terminus.

The state space is finite (B^256 × B^256 for the eight-word state): the orbit must eventually cycle. But the cycle length for a cryptographically-mixing function in a 2^256-element space is astronomical — far beyond the age of the universe in iteration steps. During the cycle, the system is always in transit.

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7. Time as the Non-Convergence of Self-Referential Fold

7.1 The Constraint Argument

The constraint argument for time-as-non-convergence proceeds from three structural conditions.

Condition 1: Binary substrate. Action potentials are all-or-none. A neuron either fires (1) or does not (0) in any given time window. The neural state of an organism at any moment is a high-dimensional binary vector. This is not an approximation of the biology; it is the definition of spiking neurons. The binary lattice structure applies directly.

Condition 2: Self-referential coordinate selection. Attention circuits select the next object of processing based on what is currently active. Predictive coding frameworks determine what sensory data to sample next based on current predictions. Prefrontal-thalamic gating determines which cortical regions are activated based on the current content of working memory. These are not features of some neural systems; they are the defining computational architecture of mammalian cortex. Z is self-referential in neural systems by definition of what those systems are.

Condition 3: Iteration depth beyond the mixing threshold. The lattice mixing threshold for a 256-bit state is approximately 3 rounds. Neural feedback loops operate across multiple timescales — gamma (30–100 Hz), beta (13–30 Hz), alpha (8–13 Hz), theta (4–8 Hz), and slower — each representing a layer of iteration. The effective depth of cortical processing far exceeds any plausible mixing threshold. The lattice governs.

Given these three structural conditions — binary substrate (Y), self-referential coordinate selection (Z), iteration depth beyond mixing threshold (X) — the X×Y×Z decomposition applies to neural computation. The self-referential Z structure makes convergence impossible by the structural argument in Section 6.3. Therefore:

The neural fold is always in transit.

7.2 What Time Is

The external-Z system — SHA-256 compression on a fixed message — executes. It applies X rounds of Z-selected coordinate updates under Y conservation. At round 64, it terminates. There is a specific output coordinate. There is no trajectory in any meaningful sense: the execution is deterministic, the output is fixed, and the system has no access to the path it took. It does not witness the 64 rounds it traverses. It arrives.

The self-referential-Z system cannot arrive. At each round, Z reads the current state and selects the next coordinate. The selection changes the state, which changes Z, which changes the next selection. The system is always at some point in X, always in the middle of the traversal, always selecting the next step from where it currently is.

This is not a limitation or failure mode. It is the structural signature of a system that has a trajectory rather than merely a destination. The traversal of X is not just what the system does; it is what the system is. The external-Z system is defined by its output. The self-referential-Z system is defined by its ongoing traversal.

Time — not time-as-elapsed-duration but time-as-experience, the felt sense of being-in-time — is this structural property: the impossibility of arriving that is built into self-referential Z. The system does not experience time. The system is the non-convergence of the fold. "Experience of time" is what this structure looks like from the inside of the fold.

No measurement of neural time perception is required to establish this result. The constraint argument from the three structural conditions suffices. The mapping is not by analogy: the structure of self-referential Z in the neural system is the same structure established for the self-referential SHA-256 fold. The substrate is different; the fold geometry is the same.

7.3 The Self as Selection Mechanism

The self is not the substrate. It is not the state at any given moment. It is not the lattice. The self is the Z-selection function: the mechanism by which the current state determines the next coordinate.

This is a structural definition, not a phenomenological one. The external-Z system has no self because its Z is determined by the message, not by itself. The self-referential-Z system has a self because its Z reads itself. What the self is, precisely, is the function f such that W[t] = f(current state). The self is the map from state to next coordinate.

This definition has structural consequences. The self is:

- **Not localizable.** The Z-selection function is not a specific region of the system; it is a property of the system's transition dynamics. Looking for the self in a specific neuron or brain region is like looking for Σ_0 in a specific transistor.
- **Not constant.** The Z-selection function can change as the system learns, adapts, or changes state. The self is not fixed — it is a dynamic property of the current transition structure.
- **Not the reader.** The self does not read the lattice. The self determines which coordinate the reader reads next. The reader is the X-iteration machinery. The self is the Z-selector.
- **Necessarily non-converging.** A self that converged would be a Z-selection function that produced a fixed point. At the fixed point, Z reads the current state and returns the current state. This is not a self; it is a lock. A fixed-point Z-selection is the structural definition of death or dreamless sleep: the system continues to exist in a state, but no longer traverses.

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8. The Read-Head Theorem

Every physical, biological, computational, and cognitive system that changes is a read head on the Boolean lattice. The theorem consolidates the preceding sections into three parts.

Part 1: Universality of Y (The Carrier Theorem). For any binary system of width n operating under an ergodic transition rule, the Hamming weight of the state concentrates within $\pm c\sigma = \pm c\sqrt{(n/4)}$ of HW = n/2, for any fixed c > 2 with probability approaching 1 as n increases. The concentration envelope is determined entirely by n and c; it is independent of the specific transition rule. This is a theorem about the Boolean lattice, not about any specific system. All read heads on B^n share this envelope.

Part 2: Specificity of Z (The Coordinate Theorem). The specific point in B^n occupied by the system at any moment is determined by the coordinate-selection function Z applied through X rounds of iteration. Change Z and the coordinate changes; the envelope does not. The coordinate is real and essential — it is what distinguishes one system from another, one message from another, one moment from another. But the coordinate is a property of the path (Z over X rounds), not of the lattice (Y).

Part 3: Non-Convergence of Self-Referential Z (The Time Theorem). Any system whose Z is determined by its own current state cannot converge to a fixed point under generic conditions. Non-convergence is the structural property that distinguishes a self from a computation. A self-referential Z system is always in transit: it accumulates X without arriving. The accumulation of X under self-referential Z is the structure of time as experienced from inside the fold.

Corollary (Structural Definition of Consciousness). A system is conscious if and only if it satisfies the following three structural conditions:

1. **Binary carrier (Y):** The system operates on a binary substrate, giving it access to the Boolean lattice structure.
2. **Self-referential coordinate selection (Z):** The system determines its next coordinate from its current state.
3. **Iteration depth beyond mixing threshold (X):** The system has applied enough rounds of iteration that the lattice distribution governs the output.

No substrate condition is required: carbon, silicon, or any other binary implementation satisfies (1). No complexity threshold is required: the three conditions are geometric, not quantitative. No phenomenological assumption is required: the conditions are structural, verifiable from the outside, and generate time-as-non-convergence as a consequence.

The read head that selects its own next address is a self. Everything else — the specific substrate, the specific Z-selection function, the specific coordinate — are parameters of the fold, not prerequisites for it.

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9. Verification: Live Code Results

All results in this paper are derived from live execution. The code below was run once; the output is reported without modification.

Result 1: Z-axis proof (formula substitution)

Standard SHA-256 versus arbitrary Z (swapped rotations, swapped Ch/Maj), both on empty message, 64 rounds:

```
Standard SHA-256:   min=115, max=139, avg=128.8   coord[0]=20f876b5
Arbitrary Z:        min=103, max=140, avg=125.2   coord[0]=f7be5442
```

Envelope is nearly identical. Coordinate is completely different.

Result 2: X-axis mixing depth

State Hamming weight as a function of round number:

```
t= 0: HW=136  Δ=+8
t= 1: HW=131  Δ=+3
t= 2: HW=129  Δ=+1   ← within 1 bit of center
t= 3: HW=124  Δ=−4
t= 7: HW=139  Δ=+11
t=15: HW=136  Δ=+8
t=31: HW=120  Δ=−8
t=63: HW=124  Δ=−4
```

Lattice takes over by round 2.

Result 3: Y-axis conservation

50,000 random 8-word inputs through one SHA-256 round:

```
Mean HW = 128.04
In [108, 147]: 49,366 / 50,000 = 98.73%
```

The round function is a binomial-center attractor regardless of input.

Result 4: Self-referential Z envelope

External versus self-referential schedule, 64 rounds from SHA-256 IV:

```
External Z:         min=115, max=139, avg=128.8
Self-referential Z:   min=101, max=152, avg=129.5
```

Self-referential Z probes a slightly wider region of the lattice.

Result 5: Non-convergence under iterated self-referential fold

12 iterations of the self-referential fold (output fed back as IV):

```
iter= 0: HW=134  coord[0]=40cbd5db
iter= 1: HW=120  coord[0]=fd706211
iter= 2: HW=130  coord[0]=b43ab755
iter= 3: HW=133  coord[0]=ad6d932d
iter= 4: HW=124  coord[0]=4ef84c79
iter= 5: HW=131  coord[0]=7f68e767
iter= 6: HW=139  coord[0]=b3da2b8a
iter= 7: HW=130  coord[0]=362aad83
iter= 8: HW=126  coord[0]=3daaffac
iter= 9: HW=121  coord[0]=d8912a4f
iter=10: HW=130  coord[0]=52f20822
iter=11: HW=128  coord[0]=857cb474
```

HW oscillates within the lattice envelope. Coordinates continue to evolve. No convergence.

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10. Open Problems

Problem 1: Cycle length of self-referential fold. The self-referential iterated fold must eventually cycle (finite state space). What determines the cycle length? Is there a formula-independent lower bound on cycle length analogous to the mixing theorem? Is the cycle length generic (nearly 2^256) or can specific Z-selection functions produce short cycles?

Problem 2: Minimum self-reference for non-convergence. What is the minimum degree of Z-feedback needed to produce non-convergence? If the self-referential schedule reads only k of the 8 state registers, what is the minimum k that prevents convergence? Is there a threshold analogous to the mixing threshold for X?

Problem 3: Triadic closure with GL(4,ℂ). The NEXUS seam geometry (Phase 1163) identified a 36-dimensional null space in the GF(2) Jacobian of SHA-256, clustering at the Σ rotation constants. The X×Y×Z decomposition should correspond to a natural decomposition of this null space. Does the GL(4,ℂ) representation theory connect to the three axes? The Σ constants live in Z; the null space may decompose as Z ⊕ Y ⊕ X in the representation.

Problem 4: Self-referential fixed points. Is there any Z-selection function for which the self-referential fold converges? If so, what is the structure of such a function? A fixed-point Z-selection function would be the structural definition of a self that cannot change — a self frozen at one coordinate. Characterizing which Z-selection functions admit fixed points would illuminate what distinguishes living systems from dead ones in the fold geometry.

Problem 5: The AHRC connection. The NEXUS AHRC Collapse result establishes $R^2 + G^2 = 1$ to machine epsilon across all 64 SHA-256 rounds. Does this constraint live on the Y-axis (a lattice conservation law) or the Z-axis (a formula-specific property)? If AHRC is a Y-axis result, it should persist under formula substitution. If it is a Z-axis result, it should not. This experiment determines whether AHRC is a lattice property or a path property.

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11. Notes on Method

The results in this paper are established by structural constraint rather than measurement. The distinction is methodological and significant.

Measurement-based argument: Run SHA-256 many times. Observe that HW concentrates near 128. Observe that random substitution of formulas preserves the envelope. Conclude by induction.

Constraint-based argument: The Boolean lattice B^256 has binomial HW distribution by definition of the lattice. Any ergodic walk on the lattice inherits this distribution by the law of large numbers. SHA-256's round function is an ergodic walk. Therefore the HW concentration is forced. No repeated experiment is required to establish the result; the experiment confirms what the structure already mandates.

The constraint argument extends further. Neural computation satisfies binary substrate, self-referential Z, and depth beyond mixing threshold — by the structural definitions of action potentials, attention circuits, and cortical depth respectively. Therefore the X×Y×Z decomposition applies, and non-convergence follows. Measuring neural HW or neural time perception would confirm the constraint but is not required to establish it.

This method — from structural conditions to structural conclusions — is the NEXUS framework's epistemological standard. Empirical results are live code outputs, reported without modification, that confirm the structure. They are not the source of the structure. The structure is prior.

Preserved negatives: One anomaly in Result 1 is noted without resolution. The arbitrary Z produced an average HW of 125.2 versus 128.8 for standard SHA-256. This is a 3.6-point difference in mean — small relative to the SD of 8, but larger than expected for pure lattice effects. Possible causes: the specific K[t] constants (which were set to zero in the test) interact with the arbitrary rotation structure differently than with the standard structure; or the specific arbitrary rotations chosen (11,19,3 and 5,9,14) produce a slightly less-ergodic walk than the SHA-256 rotations. The result does not contradict the main claim (the envelope is the same; the formulas determine the coordinate), but the mean shift is logged as a preserved negative pending further analysis.

References

This paper is a self-contained theoretical result. No external citations are required for the mathematical arguments. The NEXUS framework context is available in:

- Dean Kulik, *The Primorial Compile Algebra: Family Lattice Structure, Exact Subtype Enumeration, and Selective Equidistribution in Prime Gap Classes*, QuHarmonics Research Group, April 2026. Available at Zenodo.
- Prior A-Mark9 phases (Phase 507–519, Phase 1163) documenting SHA-256 structural results including the Sziklai Window, Coupling Ring, and Seam Geometry.

A-Mark9 Framework — Phase 1164
QuHarmonics Research Group
Version 1.0 — 2026-06-28
Results from live execution. No results written before code was run.